OOP Relationships Brief

IS-A Relationship (Inheritance)

The IS-A relationship is used in inheritance. It means that one class is a type of another class. In this case, the child class inherits the properties and methods from the parent class and can reuse its functionality.

If the sentence "A is a B" makes sense, then this is an IS-A relationship. For example: Dog is an Animal.

This relationship helps in code reusability and building a clear class hierarchy.

HAS-A Relationship (Composition)

The HAS-A relationship means that a class contains an object of another class as part of it.

Instead of inheriting from another class, the class creates an object from it and uses it when needed.

If the sentence "A has a B" makes sense, then this is a HAS-A relationship. For example: Car has an Engine.

This is useful when a class needs another class to complete its task without using inheritance.

Parent = new Child(); (Upcasting)

This statement is called Upcasting.

It happens when an object of the child class is assigned to a reference of the parent class like this:

```
Animal* a = new Dog();
```

Here, the object is created from the Dog class, but the reference type is Animal.

This is mainly used to apply polymorphism, where one parent reference can refer to different child objects.